SOL v3.7–v3.8 — Scientific Data and Asset Audit

Inventory, sources, licences, resolutions, coordinate conventions, epochs, processing, uncertainties, and what is observed vs reconstructed

2026-10-11

1. Asset inventory (shipped runtime files)

`npm run assets` (tools/validate/assets.mjs) checks every file under `public/` against a provenance rule and records the file's size, SHA-256, licence, commercial status, attribution, coverage and truth label. These go into `docs/assets/asset-manifest.json`. Final release check: **1,513 assets, 0 unmatched, 0 exact duplicates, 3 non-commercial**.

Group	Files	Size	Main sources
`assets/terrain`	1,314	109.0 MB	LOLA/LRO + SELENE (Moon 118 m), MOLA/HRSC (Mars 231 m) tile pyramids
`assets/tex`	66	160.7 MB	USGS Astrogeology controlled mosaics, NASA/JPL, Solar System Scope (CC BY 4.0), ESA Gaia sky
`assets/data`	23	41.4 MB	JPL Horizons sampled ephemerides, satellite elements and osculating states, JPL SBDB small bodies
`assets/models`	14	17.3 MB	NASA 3D Resources, radar shape models, PDS SBN Gaskell models (Itokawa, Phoebe)
`assets/catalogs`	90	13.5 MB	GCNS (Gaia EDR3/DR3) + Hipparcos-2 — 85 SHA-256-verified sector shards + named-star rows
`assets/rings`	2	0.1 MB	Voyager 2 PPS δ Sco occultation (PDS Ring-Moon Systems Node)
fonts	4	0.1 MB	SIL OFL 1.1

Licences (file counts):

Licence	Files
Public domain (USGS/NASA/JPL/PDS)	≈ 1,380
Gaia CC BY-SA 3.0 IGO plus Hipparcos (ESA, credit)	90
CC BY 4.0	17
CC BY-SA 4.0	2
SIL OFL	4
Personal / non-commercial	3 (`ariel.jpg`, `oberon.jpg`, `titania.jpg`)

The three non-commercial Uranian-moon maps are the only ones left from the original 31 (AST-003). Replacing them needs USGS-controlled Voyager mosaics at an equivalent resolution, which are not in this repository.

Truth labels (file counts):

Truth label	Files
observed altimetry	1,314
observed positions with derived distances	90
observed luminance with inferred colour	20
observed plus artistically completed (giants, representative)	17
observed radar or stereo shapes	11 + 2
derived ephemerides	10

2. Data added or changed in v3.7.1 / v3.8

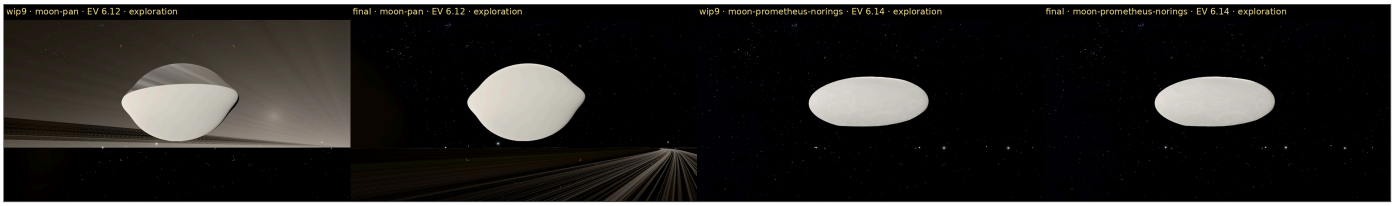
Dataset	File(s)	Source & processing	Epoch / frame	Uncertainty	Status
Gaia DR3 nearby refinement	`catalogs/stars/shards/*`, `named-rows.*`, `docs/inventory/gaia-nearby-refinement-v38.csv`, query `tools/ingest/queries/gaia_dr3_parallax_gt50mas.csv`	ADQL on gaiadr3.gaia_source (ϖ > 50 mas, ϖ/σ > 10); 2,534 GCNS rows replaced from DR3 parallax (GCNS distances quantised at 0.01 pc)	J2016.0, ICRS → ecliptic J2000	σd/d per row	OBSERVED / DERIVED
Nearby navigable stars	`catalogs/stars/named.json`	9 additional named stars incl. Proxima (1.30197 pc)	J2016.0	per row	OBSERVED

Dataset	File(s)	Source & processing	Epoch / frame	Uncertainty	Status
Photometric band (DEF-38-013)	`src/galaxy/photometry.ts`	Rows are Gaia G (Hipparcos Hp for Hipparcos-only rows); converted to Johnson V by Riello et al. 2021 Table C.2 (σ 0.030 mag); BP−RP recovered from the ingest's Teff fit	—	±0.03 (relation) + Teff-fit scatter; Proxima −0.04 vs SIMBAD	DERIVED
Uranian eccentric rings	`src/data/rings.ts`	French et al. 1988 (PDS kit `ura_rings` elements), B1950 node	epoch 1977-03-10 (French 1988)	per element	MEASURED
Saturn ring colour	`tex/saturn_ring_colour_1024.rgba`	box-smoothed (~300 km) from `saturn_ring.png` by `tools/ingest/ring_colour_smooth.py`	—	colour only (τ from PPS)	DERIVED
Itokawa / Phoebe shapes	`models/itokawa.glb`, `models/phoebe.glb`	Gaskell 2008 / 2012 ver64q plate models (PDS SBN), body-fixed km preserved; `tools/ingest/pds_shape_to_glb.py`	body-fixed	per model	OBSERVED shape
Minor-moon properties	`src/data/smallMoons.ts`, `docs/inventory/moons-v371.json`	measured triaxial dimensions, pV, colour indices, rotation for ≈ 36 moons (Thomas 1998; Thomas & Helfenstein 2020; Verbiscer 2018; Grav 2015; Karkoschka 2001/03; Weaver 2016), class priors otherwise	—	per source	MEASURED / REPRESENTATIVE
Minor-moon sizes (DEF-38-008)	`src/data/satSizes.json`, `tools/ingest/sat_sizes.py`, inputs in `tools/ingest/queries/satsize/`	JPL Horizons object data (459 of 460 NAIF codes fetched 2026-10-11: radius and V(1,0) where JPL publishes them); otherwise MPC absolute magnitude H (Wikipedia moon lists, whose H column cites MPC-NatSats); radius = 1329 km/√pV·10^(−H/5)/2 with the moon's own pV	—	H ±0.2–0.5; albedo prior factor ~2 → size ±40 %	ESTIMATED (labelled in the inspector)
Inner-moon orbits (DEF-38-009)	`src/astro/system.ts`	JPL mean elements (Laplace frame, apsidal/nodal precession) with the mean longitude anchored to the Horizons state at its epoch, for moons inside 10 planetary radii	Horizons epoch	phase as Horizons; radius = geometric a	DERIVED
Milky Way model	`src/galaxy/milkyWayModel.ts`	Reid et al. 2019 arms; Bland-Hawthorn & Gerhard 2016 discs and bar; Drimmel & Spergel 2001 dust; GRAVITY 2018/2019 R0 and Sgr A*	Galactocentric (Astropy v4.0 conventions: R0 8.122 kpc, z☉ 20.8 pc)	per parameter (σ in file)	INFERRED (model)
Deep-sky reference objects	`src/data/deepSky.ts`	9 objects, ICRS centre, distance with citation, evidence class	ICRS	per object	OBSERVED position; NOT rendered

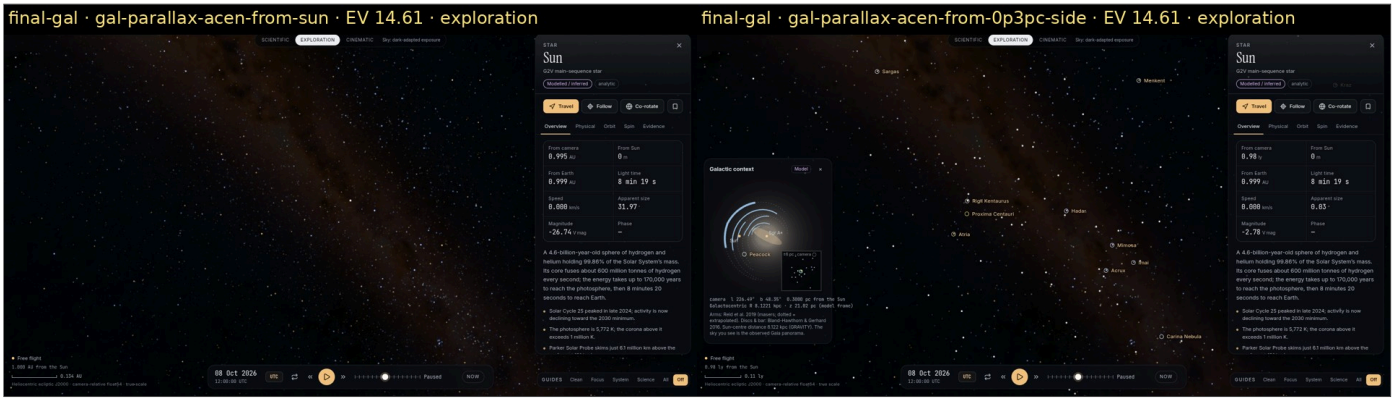
What the new data looks like in the renderer (final build, software WebGPU):



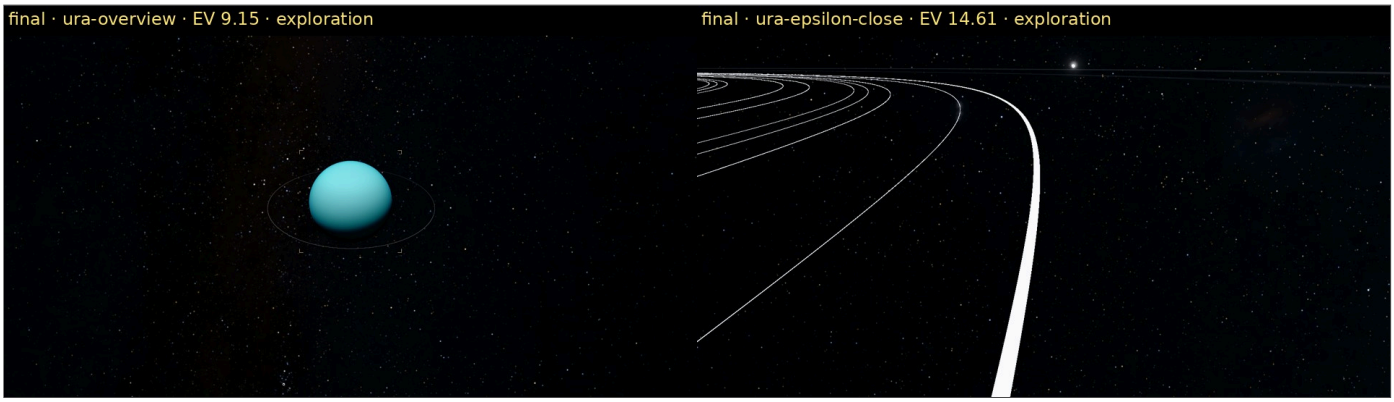
Minor moons with measured or JPL/MPC-estimated sizes and measured or class-prior albedo and colour (same view template, 3.4 R, 25° phase). Phoebe uses the observed Gaskell shape; Sycorax and Kiviuq are now drawn at their H-derived sizes, where v3.7.0 had 2.5 km placeholders.



Pan before and after the J2 inner-moon orbit correction (Pan was 500 km outside the Encke gap, so the A ring showed through it).



Gaia DR3 nearby-star positions in use: the α Centauri field from the Sun and from 0.3 pc sideways (true 3-D positions, parallax-derived distances).



Uranus with the French et al. (1988) eccentric ring elements; ε ring close-up.

3. Coordinate conventions and epochs

- **Solar-System frame.** Heliocentric ecliptic J2000, in km, held in float64 on the CPU. Rendering is camera-relative in float32.
- **Transforms.** ICRS ↔ ecliptic uses the IAU 2006 obliquity at J2000. ICRS → Galactic uses the Hipparcos (ESA 1997) rotation matrix. Galactocentric follows Astropy v4.0 conventions (no roll). All are in `src/contracts/frames.ts`, with tests: Sgr A* maps to l = 359.94°, b = −0.05°, and the NGP maps to b = 90°.
- **Stars.** Positions are given at J2016.0. Linear space motion uses proper motion, plus radial velocity where one exists (flag bit 0); otherwise motion is tangential only, and this is labelled.
- **Time.** The simulation clock is TT; the UI shows UTC, TT or TDB. Horizons fixtures are TDB, and TDB − TT is under 2 ms.

4. Observed versus reconstructed (evidence tiers)

- **O1, observed.** Planets and major moons use controlled global mosaics. Moon and Mars terrain uses altimetry. Itokawa, Phoebe and the NASA 3D Resources bodies use their shape models.
- **O2, partial.** Some maps have no data in places (noData masks are flagged, not filled as terrain). The Uranian moons' southern hemispheres were Voyager-only.
- **O3, sparse.** About 36 small moons have measured dimensions, albedo and colour. Their surface detail is a deterministic reconstruction, labelled REPRESENTATIVE.
- **O4, unresolved.** About 420 minor moons and the TNOs, rendered as class-prior families. Their sizes are ESTIMATED from H.
- **Galaxy.** A model only; the sky background is the observed 2-D Gaia panorama.

5. Validation performed

- **Ephemerides.** Horizons fixtures (`tests/astro/ephemeris.test.ts`), with tolerances fixed before comparison. The new inner-moon test keeps Pan inside the Encke gap and Daphnis inside the Keeler gap over 2020–2032.
- **Rings.** Uranian apsidal precession matches the J2 secular rate to < 3 %. Saturn's PPS regional τ is checked.
- **Photometry.**
  - Sun-like G − V = −0.15.

    - Proxima M_V = 15.6 against SIMBAD's 15.56.
- **Sizes.**
    - Sycorax 50–90 km radius (Herschel D = 165 ± 36 km, Lellouch et al. 2013; H = 8.2 at pV = 0.065 gives r ≈ 60 km).
    - Siarnaq 12–25 km; Thrymr 2–6 km; Elara 40 km (JPL/NEOWISE).
- **Milky Way.** The published Reid 2019 tangencies are reproduced; W3(OH) lies within one arm width of the Perseus arm.

6. Known data limitations (not hidden)

- **SCI-005.** `smallbodies.bin` elements are rounded to about 4 significant figures (positions off by up to 10^5 km). The full-precision SBDB fetch on promotion exists only for selected bodies.
- **SCI-004.** Mimas' interpolation error is 166 km.
- **SCI-008.** Unknown spin axes are placeholders, labelled as such.
- **Sizes.** The ~380 H-based sizes depend on the class albedo prior. A factor-2 albedo error changes a size by 41 %.
- **Hipparcos rows.** Hp is kept as approximately V ($|Hp - V| \lesssim 0.2$ mag for $B - V < 1.5$).
- **Ring material.** Particle albedos for Chariklo, Haumea and Quaoar are unknown and inferred.
- **Neptune's arcs.** Their absolute longitude is REPRESENTATIVE.

7. Missing datasets (needed for the next quality step)

- **Cassini ISS / Galileo SSI controlled mosaics** for the small moons: Prometheus, Pandora, Janus, Epimetheus, Hyperion (partial), Helene, Telesto, Calypso, Amalthea and Thebe.
- **Public-domain replacements** for the three non-commercial Uranian-moon maps.
- **Gaia DR3 bulk HEALPix tiles** beyond 100 pc, for the v3.9 Galaxy foundation.
- **Higher-resolution terrain.** LROC NAC/WAC and CTX/HiRISE regional mosaics for the Moon/Mars imagery, which is coarser than the relief.